# Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

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School of Electrical and Electronic Engineering, University College Dublin

**Application: Robotics Research Engineer, CEREBRIS Project (Ref. 020149)**

Dear Hiring Committee,

I am applying for the Robotics Research Engineer role on the CEREBRIS project. I am finishing my PhD in Mechatronics and Robotics at Aalborg University (thesis expected to be finished in March), and almost all my work so far has been robots that physically touch and move people. Building a robotic stimulator that helps measure how sensation comes back after a stroke is exactly the kind of work I want to keep doing.

In my PhD I designed a hybrid shoulder exoskeleton where a spring carries most of the arm load and a small 6 Nm motor adds the rest. I did the design in SolidWorks, 3D printed and rebuilt the parts many times, and integrated the motor, encoder and force sensors myself. It cut energy use by more than 25% and won the Best Paper Award at IFToMM ISRM 2026. I also worked with a 5-bar planar parallel robot that interacts with the human arm. I estimated the force at its tip from the variable-stiffness actuators, within 10% RMS error of a force sensor, and built an Arduino-based trigger to sync the robot's motion with Qualisys motion capture for arm tests with 6 participants.

I am used to working with people in the loop. I collected EMG and motion data from 12 participants in our exoskeleton study, and I know how much care it takes to make a session safe, repeatable and comfortable for the person in the device. My first rehab robot was an upper-limb exoskeleton in my Bachelor's, where I did the CAD, machining and assembly. In my Master's I worked on pneumatic artificial muscles and designed the outer PID loop of a gain-scheduled controller in MATLAB/Simulink (IEEE CONECCT 2022).

I have not worked on sensory testing after stroke yet, and I would be keen to learn that side from the clinical partners. What I can bring is hands-on robot hardware, sensor integration and careful human testing. I also have a Diploma in Fine Arts and I make the sketches and visuals for our papers and demos, which helps when explaining a device to clinicians or patients.

Thank you for considering my application. My portfolio is at arunabha22.github.io/arunabha-majumder.

Kind regards,
Arunabha Majumder